

Network Performance – Delay

Module 2.2

READING

- Reading: 1.4

Learning Goals

Delay

- List and define the types of delay and how they contribute to over-all delay
- Calculate the end-to-end delay in a network.
- Perform bottleneck analysis on a path
- Compute traffic intensity and relate traffic intensity to queuing delays
- Calculate link utilization
- Use the formula of Average Delay = $S/(1-U)$ where U is the network utilization and S is the average service time for a single packet (this formula only applies to randomly arriving packets)

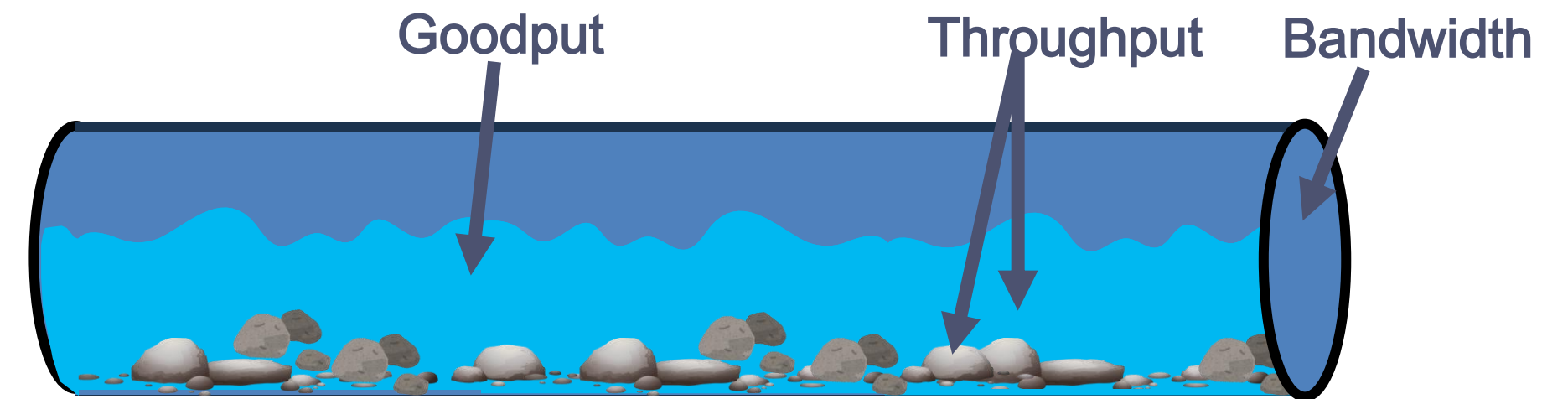
Recap

- Bandwidth
- Throughput
- Goodput

Rates
measured in
bits per
second

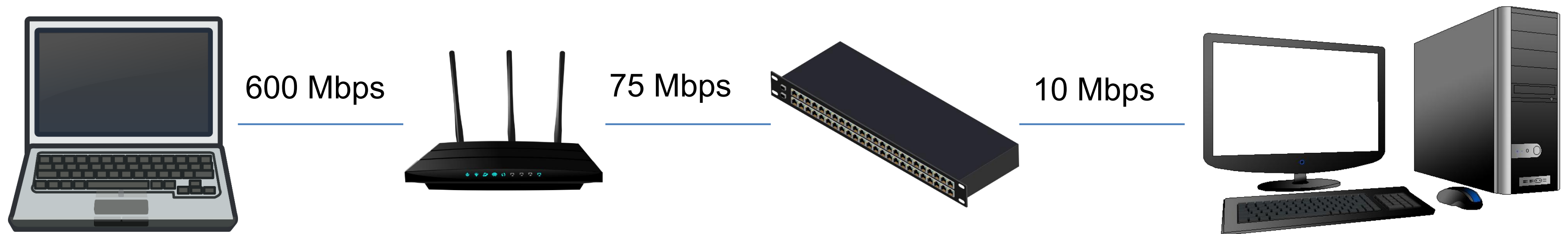
- Latency
- RTT (Round Trip Time)
- Jitter

Times
measured in
seconds



Bottlenecks

- What is the maximum throughput (possible bandwidth) between two nodes connected by a network?
 - Can traffic flow at maximum bandwidth in all links?

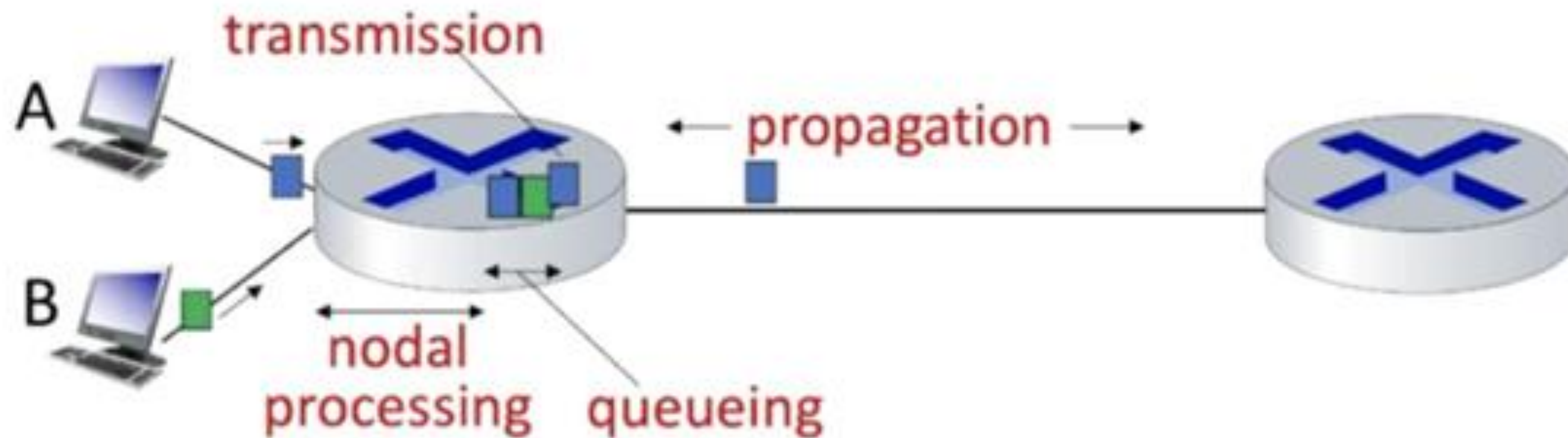


Types of Delay

- *Processing delay*: examine packet to decide where to direct it
- *Queueing delay*: waiting time to get access to the link
- *Transmission delay*: time to actually write the packet onto the medium
- *Propagation delay*: time spent to move each bit from source to destination on the transmission medium
- *End-to-end delay*: sum of all sources of delay

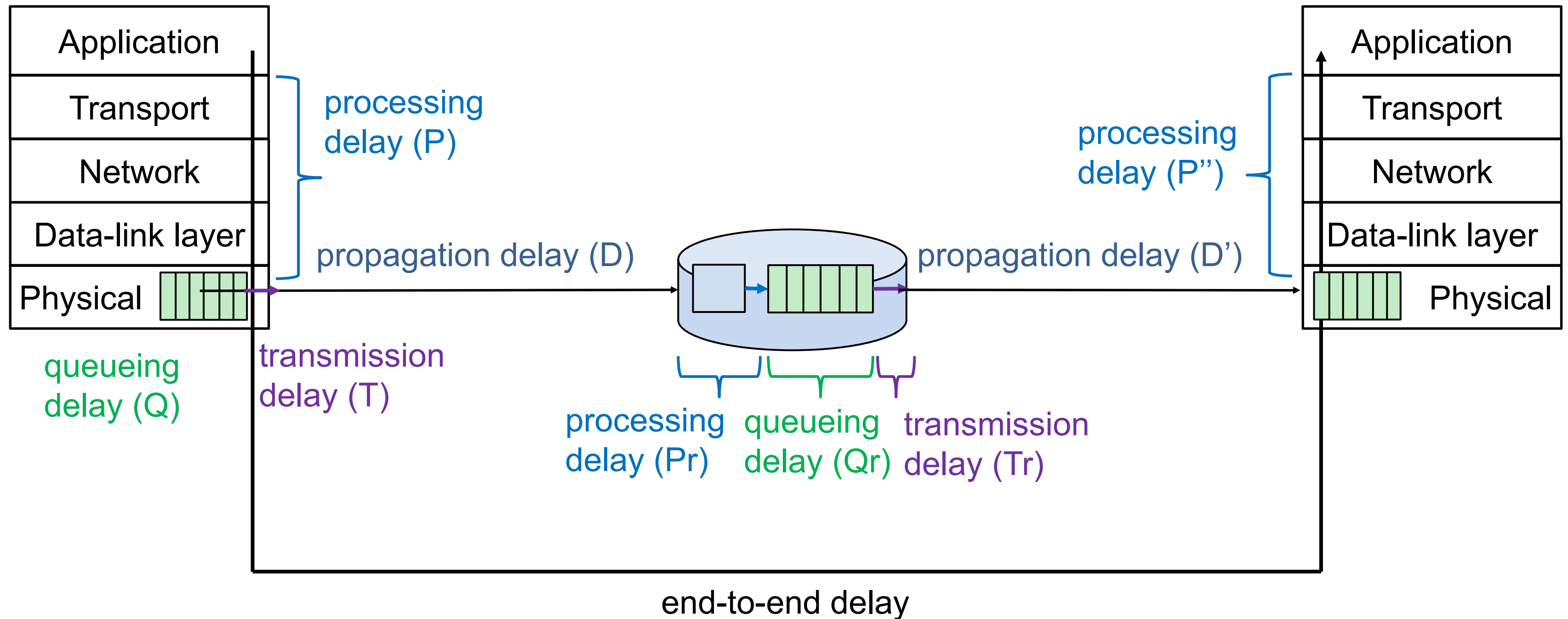


Types of Delay



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

Types of Delay



Classification of Delay: fixed vs variable

- | | | |
|-----------------------------|--------------------|--------------------|
| • <i>Processing delay</i> | | fixed or variable? |
| • <i>Queueing delay</i> | | fixed or variable? |
| • <i>Transmission delay</i> | } <i>same link</i> | fixed or variable? |
| • <i>Propagation delay</i> | | fixed or variable? |
| • <i>End-to-end delay</i> | | fixed or variable? |

Suppose ...

You are sending the same sized packets to the same destination over and over again.

- *Processing delay.* ?
- *Queueing delay.* ?
- *Transmission delay.* ?
- *Propagation delay.* ?
- *End-to-end delay.* ?

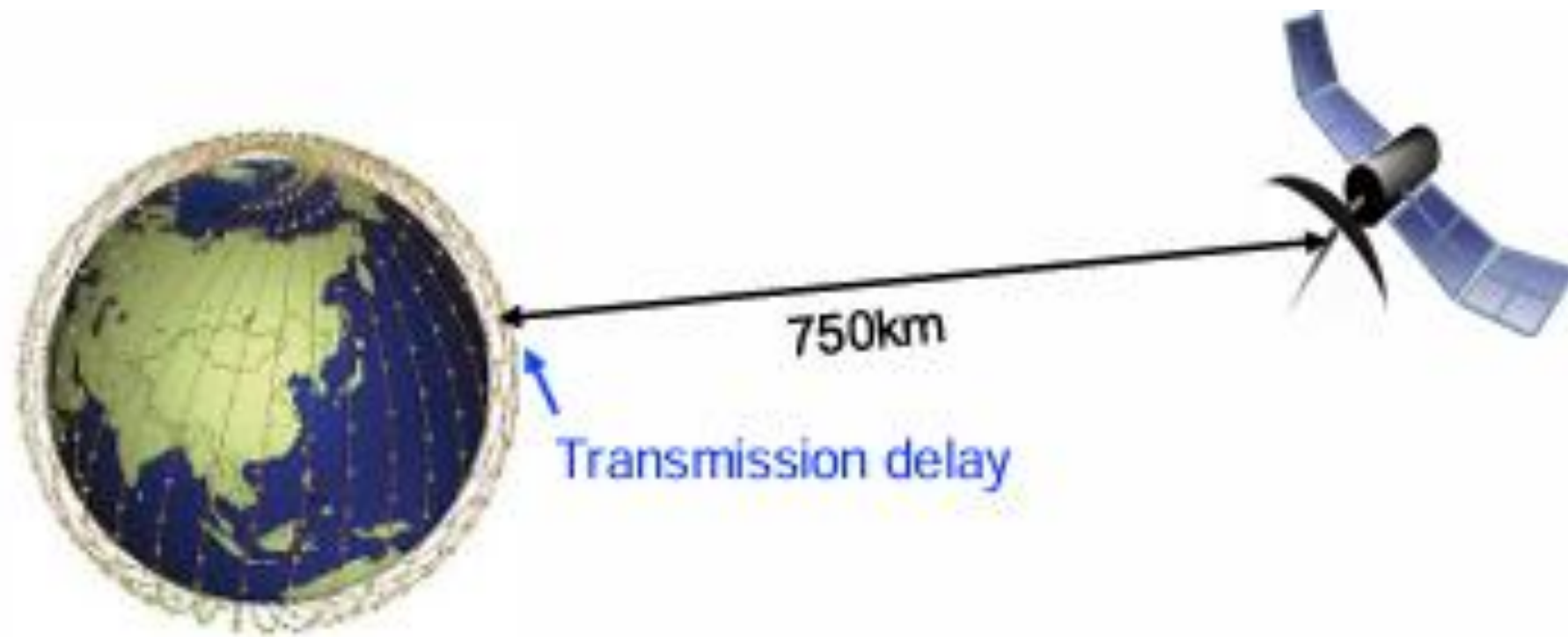
Delay Calculations – Propagation

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. **What is the two-way propagation delay?**



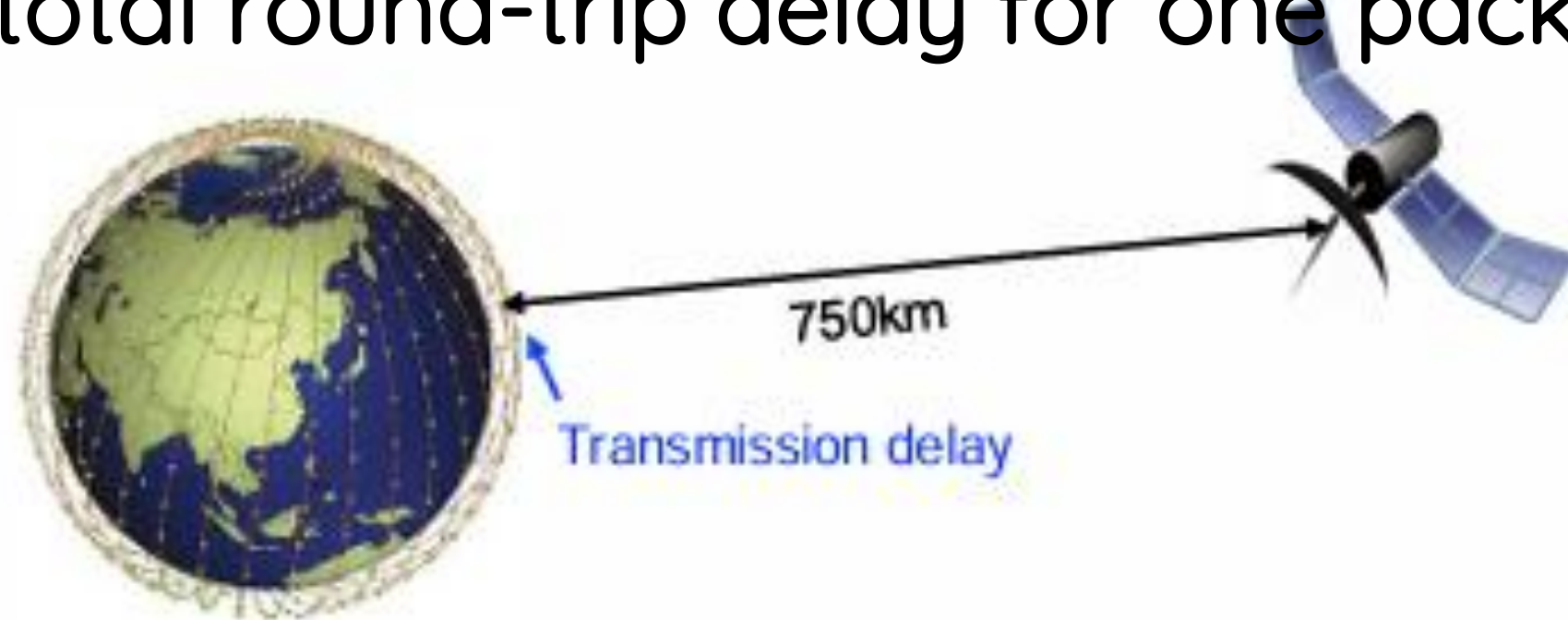
Delay Calculations – Transmission

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. The network has a transfer rate (bandwidth) of 100Mbps. **What is the transmission delay (one-way)?**



Delay Calculations – End-to-end

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume the transmission delay for the ACK is 0ms, and that the processing and queueing delays are 0. **What is the total round-trip delay for one packet?**



More Calculations – Throughput

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume that the processing and queueing delays are 0. Assume also that you can transmit packets back-to-back.

What is the throughput?

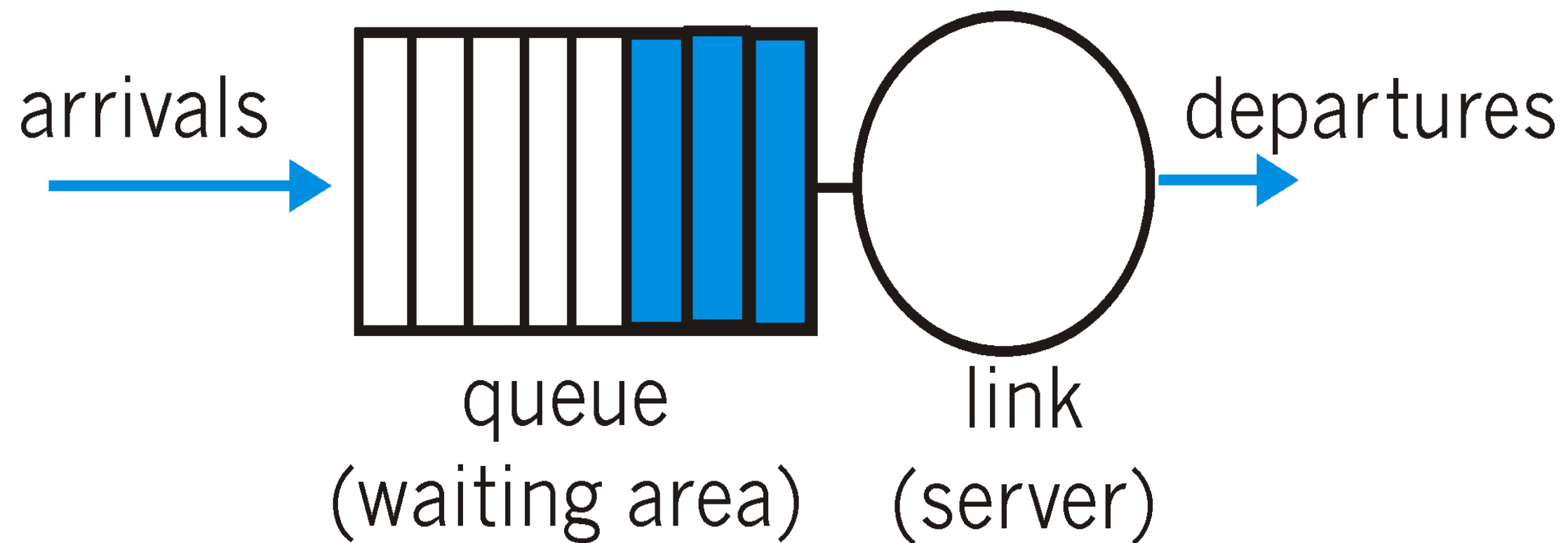
More Calculations – Throughput

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume that the processing and queueing delays are 0. Assume also that you don't transmit the $n+1^{\text{th}}$ packet until you get a (very short) ACK for the n^{th} packet (assume 0ms transmission delay for the ACK). **What is the throughput?**

Traffic Intensity

- How much data is arriving at the router?
- How much data can a router handle?
 - At what rate can the router forward data out?
- Queueing: when a router receives data faster than it can forward it

Traffic Intensity



Arrival Rate of 100 packets/second

Packets depart every 1 millisecond

Traffic Intensity Calculation

- Traffic intensity is determined by
 - Number of packets arriving per second (a)
 - Average packet size (L) in bits
 - Transmission rate: rate at which bits are disposed off per second (R)
- Traffic intensity: La/R
- Example:
 - Suppose a router is connected to a 1Mbps link. The router receives an average of 100 packets per second, averaging 500 bytes per packet. What is the traffic intensity?

Traffic Intensity Rationale

- Traffic intensity helps us understand how busy a link is
- Queueing delay is related to the intensity
 - Queueing delay is delay caused by waiting for the queue to clear
 - Packets arriving must wait for packets already there to leave

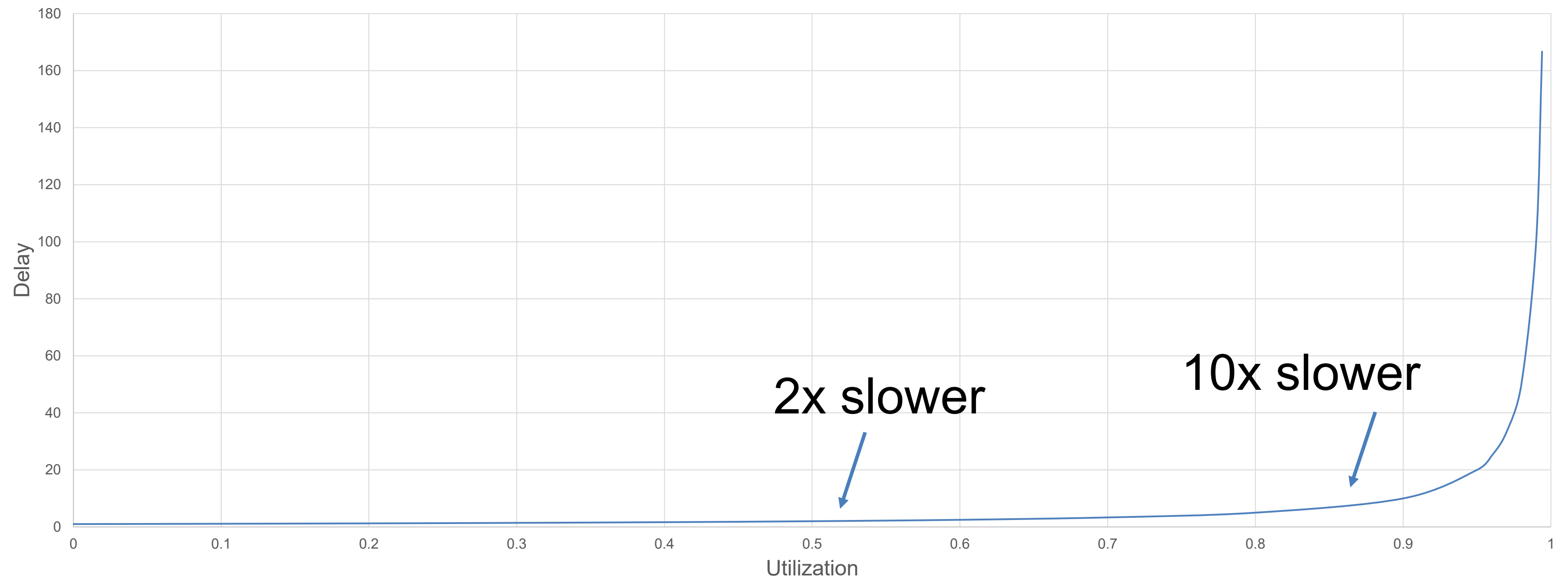
Traffic Intensity vs Queueing Delay

- Suppose λa bits/second arrive randomly for an outgoing link in a router
- Suppose that the router can transmit R bits/second
- Draw a graph of queueing delay vs traffic intensity
 - What does $\lambda a = R$ mean?
 - What does $\lambda a > R$ mean?
 - What does $\lambda a < R$ mean?
 - What does $\lambda a \ll R$ mean?

Queueing Problem

- **Packets are not spaced out evenly**
 - Spacing between packets is not deterministic
- Packets may not be served evenly
 - Link may be busy at times
 - Particularly for shared medium (e.g., radio signals)
- Higher intensity means higher probability that there is one or more packets in the queue

Delay vs Utilization



Traffic Intensity vs Queueing Delay

- Assuming packets arrive at an exponential distribution, delay is given by:

$$\text{Total Delay} = \frac{S}{1 - U}$$

- Where:
 - S is average service time when server is idle
 - U is server utilization (usually traffic intensity)
- Queueing delay is then:

$$\text{Queueing delay} = \frac{S}{1 - U} - S \text{ or } \frac{US}{1 - U}$$

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Some Observations

- Routers don't have infinite buffer space
- If packets arrive faster than they can be disposed off, they may have to be dropped
- Packets may also be corrupted in transit
 - These packets must be discarded, since their content is no longer valid
 - Even routing information (e.g., destination IP) may be corrupted

In-class Activity

- ICA22

Next Topic: Application Layer Protocols